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Motion sickness and cybersickness – Sensory mismatch

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ABSTRACT

The use of virtual reality (VR) with head-mounted displays (HMD) may cause side effects called cybersickness with symptoms comparable to those of motion sickness. In this study, we explored whether individual balance characteristics and self-reported tendency to motion sickness could be related to cybersickness vulnerability.

Healthy young people ($N = 45$) were exposed to a VR application with HMD for four minutes, standing with no support. Balance characteristics were measured before (Sensory orientation test) and during (balance platform) the VR exposure. Symptoms of cybersickness were recorded by the Simulator sickness questionnaire (SSQ). Data were analyzed for subgroups with and without a tendency to motion sickness.

The participants were negatively affected by the VR exposure: SSQ-before: 21.3 (19.5); SSQ-after: 31.8 (25.2); $p<0.01$, and 73% experienced increased discomfort. The SSQ sub-scores Nausea and Disorientation were affected, but not the sub-score for Oculomotor disturbance.

Surprisingly, the participants described discomfort already after the initial balance assessment (Sensory orientation test). Participants with a self-reported tendency to motion sickness were relatively more affected by this challenge to their sensory integration.

Increased postural instability was evident during the VR exposure, but there was a sizeable individual variance in the postural response. The study identified no individual balance characteristics which could be associated with the cybersickness vulnerability.

The adverse effect of the Sensory orientation test is a novel finding and it became a bias that diminished subgroup differences in cybersickness vulnerability.

1. Introduction

The use of virtual reality (VR) has great potential, but some users of VR may experience visually induced motion sickness, also called cybersickness [1]. Cybersickness is characterized by the onset of nausea, oculomotor disturbance, and/or disorientation while being exposed to virtual environments in a Head-mounted display (HMD), large screen, and curved screen [2]. The likelihood of experiencing these symptoms is as high as 22%–56% [3].

Visual, vestibular, and somatosensory cues are normally integrated for self-motion perception, but in VR, these cues may conflict with one another, causing cybersickness [4]. When the VR user is standing or sitting, and the VR visual illusion provides the sensory input of the person being stationary, no sensory mismatch occurs. In contrast, the sensory mismatch may elicit cybersickness when the virtual visual illusion informs the sensory system that movement and locomotion occur, while the bodily and vestibular senses reflect no movement [5]. It may be suggested that individual sensory prioritization for postural control strategies influences the vulnerability to cybersickness.

Cybersickness is a sensation very similar to motion sickness, with the difference being that physical movement is usually limited or absent during VR exposure [6]. Like cybersickness, motion sickness may be elicited by a sensory mismatch [7]. Seasickness may occur when situated in a cabin on a ship at sea in rough weather, and car sickness may be elicited when reading a book while sitting in a car in motion. In both cases, the visual information is not aligned with the vestibular and sensory-motor information about movement and locomotion. Postural instability has been linked to motion sickness [7] and cybersickness, although a large individual variation is observed [8]. When advocating for the use of VR, knowledge about individual characteristics that may influence the vulnerability to cybersickness is needed.

The present study aimed to explore the relation between individual

Abbreviations: SOT, Sensory Orientation Test; SSQ, Simulator Sickness Questionnaire; HMD, Head-mounted Display; VR , Virtual reality.

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postural control characteristics, self-reported vulnerability to motion sickness, and the occurrence of cybersickness in a VR setting with sensory conflicts. We hypothesized that those who had a known tendency to motion sickness would also be more vulnerable to cybersickness.

2. Material and methods

2.1. Participants

Healthy participants between 18 and 35 years were invited for the study through public notice. Exclusion criteria were neck pain, previously diagnosed neurological conditions, migraine, dizziness symptoms, known balance problems, persistent symptoms from a concussion, sensitivity to light, and dependence on glasses with transition lenses. The participants' general characteristics were recorded by a questionnaire on age, gender, VR-use, and self-perceived motion sickness tendency.

The local research ethics committee accepted the study, and informed consent was obtained from all participants in accordance with the Declaration of Helsinki.

2.2. Protocol

The participants were at first tested in a standardized balance evaluation protocol, the Sensory orientation test. Then they had 5–10 min of rest before being asked to answer a Simulator Sickness Questionnaire (SSQ) on symptoms and discomfort. This was regarded as the baseline discomfort score.

Subsequently, they were introduced to the VR equipment. They would put on the HMD and orient themself to the virtual environment. When they felt ready, they started a VR roller coaster ride. During the VR experience, the participants were standing on a balance platform, which measured the postural sway.

After the VR exposure, the participants removed the HMD and stepped down from the balance platform. Within a minute after the VR exposure, they began to fill out a follow-up SSQ questionnaire.

2.3. Outcome measures

The participants' individual balance characteristics were evaluated by the Sensory orientation test (SOT). which challenges the postural control in six different standardized sensory conditions in an instrumented setting (Smart Equitest, Neurocom, Natus Medical Inc.). In some conditions the visual field may follow the sway-pattern in a way that is not providing a visual reference point, and/or the support surface may follow the sway-pattern and thereby minimize the input from ground reaction forces. The output is based on the anterior-posterior sway excursion with reference to a max sway envelope of 12.5° [9]. The composite score (no unit) from the six test conditions was used for the analysis, and normative data were extracted for comparison from Pletcher et al. (2017) [9]. Additionally, the SOT was complimented by the standardized head shake tests (SOT-HS) on a fixed surface and a sway-reference surface. In these tests, the participant rotates the head from side to side at a controlled pace and amplitude, and the results are presented as the relative performance (percentage) with reference to the corresponding eyes-closed score without head shake.

The postural activity during the VR exposure was evaluated by a stable balance platform (Good Balance system, Metitur Oy), which recorded the center of pressure (COP) with a sampling rate of 50 Hz [10]. The COP trajectory was evaluated with respect to mediolateral and anterior-posterior velocities, and COP velocity moment (i.e., the mean area covered by the movement of COP during each second of the test, taking into account both the distance from the geometrical midpoint of the whole test and the speed of movement during the same period (mm^2/s) [11]. For comparison, normative values for standing undisturbed with eyes closed had previously been collected from 15 healthy students in the same age group. The closed-eyes situation was believed to be best compared to wearing an HMD, where the visual reference frame is also lacking. These data were used as norm-referenced data.

The self-reported discomfort was evaluated by the Simulator Sickness Questionnaire (SSQ). This is a 16 items questionnaire measuring discomfort within three categories: symptoms of nausea (stomach awareness, etc.), oculomotor symptoms (headache, eyestrain, etc.), and disorientation (vertigo, dizziness, etc.) [12]. The severity of discomfort is graded from 0 - 3, and the scores may be weighted and summed up into a total score of discomfort (SSQ-total). The participants responded to the questionnaire before and after the VR exposure. The SSQ was originally developed to categorize self-reported symptoms related to flight simulators [12], but self-reported symptoms are also evident after VR immersion and may still be reported after half an hour [13]. The severity of symptoms has been graded according to the use of SSQ-total in flight simulators: 0 no symptoms; <5 Negligible symptoms; 5–10 Minimal symptoms; 10–15 Significant symptoms; 15–20 Symptoms are a concern; >20 A problem simulator [14]

2.4. VR exposure

The lead investigator launched the computer program and prepared the virtual software arena following the software companyś manual. During the VR exposure, the participants wore an HMD of the brand Oculus Rift (Oculus, Facebook Technologies, LLC). They were standing with no support, but a member of the research team was ready to provide support if needed. At first, the participants were introduced to the virtual arena, and when they felt comfortable, they were exposed to a virtual ride on a rollercoaster ("Epic Roller Coaster classic"), which lasted for 3 min and 50 s [15].

2.5. Statistics

SSQ scores were weighted according to Kennedy et al. [12] and presented by means and standard deviations. As some data sets were not normally distributed, the SSQ scores recorded before and after VR exposure were compared by Wilcoxon non-parametric tests. The Mann-Whitney U test was used to evaluate SSQ-total differences between groups with and without motion sickness tendency. As both the total score and the three subscale scores were analyzed, Bonferroni corrections were used resulting in an alpha level of 0.0125.

The postural measures were compared by Mann Whitney *U* test and effect sizes by Cohen's d. Associations between the individual balance characteristics measured by the SOT and the SSQ-total scores were evaluated by Spearman's correlation coefficients.

3. Results

We included 45 participants (16 females) with a mean age of 26.5 years (SD 3.4). None of the participants were using VR frequently, but 50% had tried VR previously. A tendency to motion sickness was reported by 19.6% of the participants.

3.1. Balance characteristics

Individual balance characteristics evaluated by the Sensory Orientation Test and the Head shake tests were comparable to the norm reference values for the age group. No differences were seen between the subgroups with and without motion sickness. (Table 1).

There were increased postural corrections during the VR exposure. The values of COP mediolateral and anterior-posterior velocities and the COP velocity moment during the VR exposure were significantly higher than the norm-reference values for standing with eyes closed. The group with motion sickness tendency had smaller COP excursions, but no statistically significant differences were seen between the subgroups (Table 1).

Table 1
Balance characteristics, subgroup data, effect size, and norm reference data.

	All (N = 45)	Motion sickness (N = 9)	Non-motion sickness (N = 36)	Effect size	Norm-referenced data
Sensory orientation test					
SOT (composite score no unit)	79.0 (4.7)	79.0 (5.8)	79.0 (4.5)	*0.00*	*80 (5)*
HS SOT Fixed surface (%)	98 (3.5)	99.0 (3.8)	97.9 (3.3)	*0.33*	*N/A*
HS SOT Sway-reference surface (%)	63 (29.6)	61.1 (26.2)	63.5 (30.7)	*0.08*	*N/A*
Postural sway measures during VR-exposure					
COP Mean X Speed (mm/s)	25.4 (12.7) *	19.3 (10.7)	27.0 (12.8)	*0.62*	*4.6 (1.8)*
COP Mean Y Speed (mm/s)	48.5 (18.1) *	40.0 (15.7)	50.6 (18.3)	*0.59*	*10.3 (3.4)*
COP Velocity-moment (mm^2/s)	541.9 (413.1) *	373.5 (295.3)	583.0 (430.7)	*0.52*	*15.4 (8.3)*

Mean (sd). * $p<0.001$ comparing All and Norm-referenced data.

3.2. Self-reported discomfort

The initial SOT balance assessment, with challenges the sensory integration, initiated discomfort amongst the participants. The baseline SSQ scores, retrieved shortly after this test, were high, according to the original simulator evaluations [14]. The mean SSQ-total score was above 20, which is a score that originally indicated problems with a flight simulator (Table 2).

The participants were generally negatively affected by the VR exposure, and the SSQ-total increased by 50%. This difference was statistically significant ($p<0.0125$), and a negative difference was observed among 73% of the participants.

Concerning the sub-categories in the SSQ questionnaire, the scores were negatively affected in the Nausea in both groups, and the Disorientation and Total in the non-motion-sickness group (Table 2).

Statistically significant differences in SSQ scores were observed when comparing subgroups with and without motion sickness tendency, both in the Total score at baseline and in the Nausea and Total score after VR exposure. No significant group differences were seen for the increased SSQ scores elicited by the VR exposure.

No strong correlations were seen between the individual balance characteristics and the SSQ-total scores (Table 3).

Table 2
SSQ scores before and after Virtual reality exposure (N = 45). Subgroups organized with respect to self-reported tendency to motion sickness.

	Baseline	After VR	Difference
All (N = 45)			
Nausea	15.2 (16.6)	29.3 (22.9)	14.0 (18.4) *
Oculomotor	16.3 (16.6)	18.5 (18.4)	2.2 (14.5)
Disorientation	26.9 (25.8)	41.8 (37.4)	14.9 (24.2) *
Total score	21.3 (19.5)	31.8 (25.2)	10.6 (17.2) *
Motions sickness (N = 9)			
Nausea	26.5 (17.5)	50.9 (29.4) ¤	24.4 (19.7) *
Oculomotor	27.0 (17.0)	30.3 (24.0)	3.3 (17.4)
Disorientation	47.9 (32.0)	72.7 (42.2)	24.7 (25.8)
Total score	36.6 (21.0) ¤	54.4 (31.9) ¤	17.9 (19.3)
Non-motion sickness (N = 36)			
Nausea	12.4 (15.4)	23.8 (17.6)	11.4 (17.3) *
Oculomotor	13.7 (15.7)	15.6 (15.9)	1.9 (14.0)
Disorientation	21.7 (21.4)	34.0 (32.3)	12.4 (23.59 *
Total score	17.5 (17.4)	26.2 (20.0)	8.7 (16.5) *

Mean (sd), no units.
Exposure difference: * $p<0.0125$. Subgroup difference: ¤ $p<0.0125$.

Table 3
Correlations between balance characteristics (Sensory orientation test) and SSQ scores.

	SSQ-total Baseline	SSQ-total After VR	SSQ-total Difference
SOT (no unit)	-0.08	0.03	0.05
HS SOT Fixed surface (%)	0.04	0.05	0.07
HS SOT Sway-reference surface (%)	-0.23	-0.14	0.06

Spearman's correlation coefficient.

4. Discussion

4.1. Main observations

The SSQ baseline score following the initial balance assessment was high, indicating that the SOT and HS-SOT assessments had elicited discomfort amongst the participants. The average SSQ-total score was 21.3 (19.5), which would have been regarded as a problematic high score had it been elicited by an aviation simulator [14] Especially the participants with a tendency to motion sickness were affected by this assessment protocol, which challenges the sensory integration.

The short VR exposure, characterized by conflicting visual and vestibular/proprioceptive sensory information, elicited an increased feeling of discomfort for many of the participants. The higher SSQ scores amongst 73% of the participants may be interpreted as the occurrence of cybersickness. The SSQ scores after VR exposure were relatively higher in the motion sickness group, but the SSQ increase was not significantly different between the groups. This may, however, be ascribed to the higher baseline score in the motion sickness group, and the negative finding does not imply that there is no relation between motion sickness and cybersickness.

4.2. Postural control

The postural control was strongly affected during the VR exposure, and the COP excursions were extensive compared to reference values for people standing with closed eyes.

Postural control may be defined as the act of maintaining, achieving or restoring a state of balance during any posture, or activity [16]. The postural control is based on sensory information from three reference systems [17] (Fig. 1):

1 The exo-centric reference – where the visual input, supplemented by auditory and other exteroceptors, will provide information about the surroundings
2 The geo-centric reference – where the vestibular organs will inform about the orientation of the head's position and movement in space with reference to gravity (vertical alignment) and inertial forces during movement
3 The ego-centric reference – where the somatosensory organs will provide proprioceptive input that may register the positions and movements of the body parts and their alignment

We expected to see differences in the participants' presumptions for handling the sensory conflict in the VR exposure. To evaluate the individual characteristics for sensory prioritization, the participants' balance characteristics were tested by the SOT and the HS-SOT in the Equitest setup. However, the findings from these tests were neither related to the SSQ scores nor motion sickness tendencies. Also, no correlations between the individual SOT scores and the differences in SSQ scores were found.

We hypothesized that people vulnerable to motion sickness would be more affected by cybersickness. It has previously been shown that these

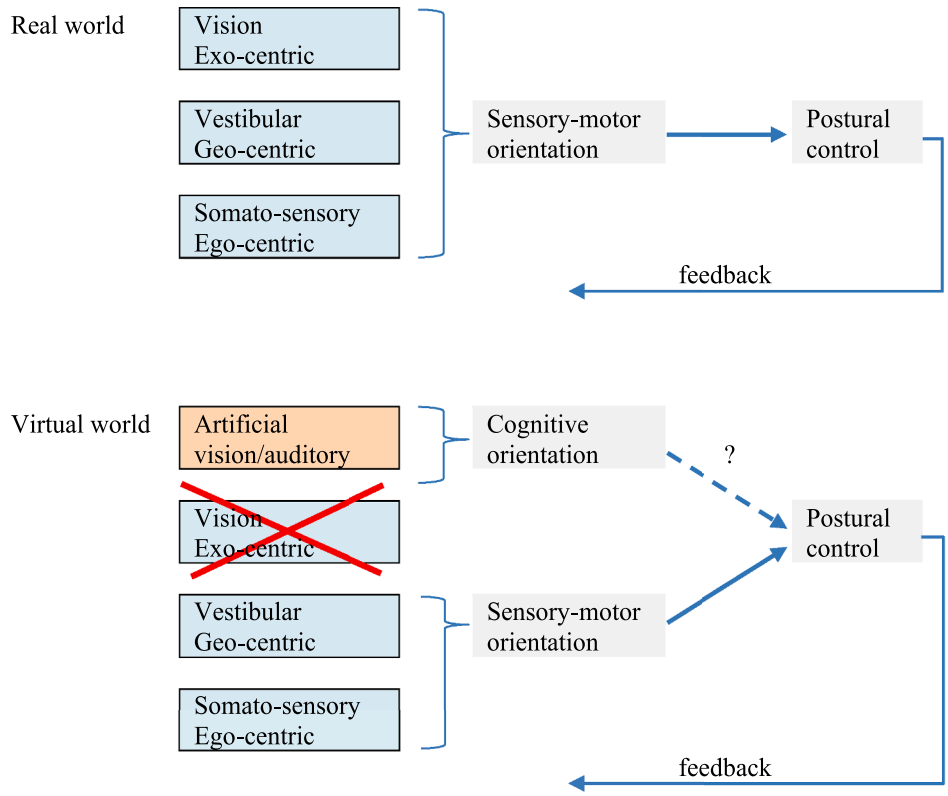

Fig. 1. The postural control depends on visual, vestibular, and somatosensory information about the exo-, geo-, and ego-centric references, from which it may be orientated. When using HMD in a VR setting, the visual (and auditory) information is replaced by artificial stimuli, facilitating a cognitive orientation that may be different from the sensory-motor orientation.

two conditions may have the same origin [18]. Although the participants with motion sickness tended to be more affected by VR, especially with respect to nausea symptoms, no statistically significant findings could support our hypothesis.

The SOT assessment procedure consists of controlled manipulation of the sensory input and guided head rotation. Apparently, the participants were strongly affected by this testing procedure. Although we allowed a "wash-out" period of 5–10 min before the baseline SSQ questionnaire was filled out, the participants had a very high baseline SSQ score. Furthermore, the subgroup with a tendency to motion sickness was significantly more affected by this assessment protocol.

This adverse effect of the SOT/HS-SOT is an observation that is sparsely described in the literature [19]. It may not come as a surprise, however, that people who are vulnerable to the sensory mismatch eliciting motions sickness would also be susceptible to the sensory discrimination challenges which are provided during the SOT assessment. With reference to Fig. 1, the sensory manipulation in SOT is not based on a substituted visual input (as in VR). The visual and proprioceptive input is, however, altered in a way that is unexpected and different from a real-world setting.

In the present study, the adverse effect of SOT became a bias for the evaluation of the impact of VR exposure. As the wash-out period after the balance assessment apparently was too short, the baseline measure became unrealistic high and diminished the differences between the two SSQ evaluations.

4.3. Visually induced motion sickness - Cybersickness

Although the increases in the SSQ scores were minor amongst many participants, the general cybersickness effect of the VR exposure seemed evident. A computer-generated visual illusion may provide the cognitive experience of a virtual environment that involves vection (Fig. 1). Vection describes the sensation of illusory self-motion in the absence of physical movement through space [6].

Sensory information is integrated to estimate the postural orientation of the body. This perception of meaningful interaction with the surroundings is based on an individual learning process through years of movement experience. An unexpected sensory mismatch may result in a new learning process for predictive coding in the brain [20]. The sensory mismatch may, however, also cause adverse symptoms. When VR elicits the symptoms, they may be termed visually induced motion sickness or cybersickness. Motion sickness is also elicited by a sensory mismatch but with a different origin [21]. In a car, the driver is fully oriented with aligned sensory input from the visual, vestibular, and somatosensory systems, while the passenger may read a book, or he/she may sit in the backseat with a poor view of the road and coming events. The passenger may be lacking feed-forward mechanisms in motion perception [22] and postural activity [23], and he/she may therefore have symptoms while the driver does not.

In the present study, the visual illusion of riding a roller coaster in the virtual environment conflicted with input from the bodily senses, registering the body in a standing position with no locomotion. When a participant experienced an increased discomfort, it may therefore be ascribed to this sensory mismatch [5]. Far from all VR applications provide the illusion of unrealistic locomotion as the virtual rollercoaster ride does. But this application was chosen in order to provoke the participants and identify individual characteristics that could be related to cybersickness symptoms.

The SSQ questionnaire evaluates cybersickness symptoms. It consists of the subcategories: Nausea (SSQ-N), Oculomotor (SSQ-O), and Disorientation (SSQ-D). In previous studies the subcategories have demonstrated that different stimuli provide different side effects: flight simulators: SSQ-*O*>SSQ-*N*>SSQ-D; seasickness: SSQ-*N*>SSQ-*O*>SSQ-D; space sickness: SSQ-*N*>SSQ-*D*>SSQ-O; cybersickness: SSQ-*D*>SSQ-

N>SSQ-O [2]. In the present study on cybersickness, we also found this relative distribution of the scores: SSQ-*D*>SSQ-*N*>SSQ-O.

4.4. Postural control during VR

During the VR exposure, the balance adjustments, registered as the COP trajectory velocities and area, were higher than the norm-referenced measures for standing with eyes closed. These figures reflect major postural corrections, which were also evident when observing the participants. Hence, the visual illusion from the VR experience highly influenced postural control.

The increased postural corrections during the VR exposure indicate that many participants were depending on the exocentric reference system and became strongly influenced by the artificial visual input. The visual dependency was not sufficiently overruled by the vestibular and proprioception systems in the sensory integration. It is well known that a visual illusion may affect postural sway and similar observations of postural instability with unexpected visual perturbation in VR have been described in other studies [24].

The observed variance in the COP measures indicates that some participants were more affected than others. It has previously been described that motion sickness may be preceded by postural instability [25], but in contrast to our expectations, we did not observe differences in postural instability during the VR exposure between the group with and without motion sickness tendency.

4.5. Study limitations

A selection bias may have compromised the external validity of the study. During the recruitment, the participants were informed about the VR exposure before enrolment. Consequently, some participants refrained from participation because of prior negative experiences with VR.

In general, the preliminary balance test negatively affected the participants, as indicated by unexpected high SSQ scores before the VR exposure. We used a 5–10-minute wash-out period after the SOT, but many participants were still affected by the test. Especially participants who considered themselves vulnerable to motion sickness had high SSQ baseline scores, and this may have diminished the measured effect of the VR exposure.

Furthermore, the SSQ questionnaire was not originally developed to register cybersickness, which tends to produce symptoms merely related to disorientation [2]. Future studies on cybersickness may prefer a new questionnaire directed toward virtual reality sickness [26] when this tool is further validated [27].

4.6. Perspectives

Head-mounted displays (HMD) are widely used to engage in virtual reality (VR) activities for individual entertainment and marketing purposes. In the health sector, VR may be used for the rehabilitation of patients and prophylactic training [28, 29]. With the widespread use of VR in society, user experience and -tolerance must be of utmost importance. Knowledge about the side effects of VR use and its potential adverse impact on postural control is relevant for the users and the manufacturers of VR soft- and hardware. Whether habituation to the artificial sensory exposure is beneficial for sensory integration is unknown. An immature brain will likely be more susceptible to manipulating sensory integration. It is worth noticing that some manufacturers of VR headsets recommend a user age limit of 12–13 years. Knowledge about the potential risks and long-term consequences of VR exposure is sparse [2]. In addition, adverse effects of VR may depend on individual characteristics and the character of sensory conflicts in VR exposure. Consequently, more research within this field is needed.

5. Conclusion

Healthy young people experienced symptoms of nausea and disorientation following a balance assessment protocol (SOT) that challenges sensory integration. People with a self-reported tendency to motion sickness were more affected by this testing protocol.

Increased discomfort (cybersickness) was experienced by 73% of the participants after a short VR exposure, characterized by conflicting visual and vestibular/proprioceptive sensory information. Postural instability was evident during the VR exposure, but there was a sizeable individual variance in the postural response to the visual illusion in the VR environment.

The study identified no individual balance characteristics which could be associated with the cybersickness vulnerability.

Source of work

Experimental study with self-reported and objective data

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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